2019

STATISTICS — HONOURS

Paper : CC-5

(Linear Algebra)

Full Marks : 50

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

1. Answer ***any ten*** questions : 1×10

(a) If A is a 10×8 real matrix with rank 8, and b is a 10×1 vector which belongs to the column space of A, then do $Ax = b$ have a unique solution?

(b) Let A be a $n \times n$ real matrix. Show that if $A^k = 0$ for some $k > 0$, then trace $(A) = 0$.

(c) Consider the system of linear equations : $(4d - 1)x + y + z = 0$; $-y + z = 0$; $(4d - 1)z = 0$. For what value of d does it have a non-trivial solution?

(d) Let A and B be $n \times n$ real matrices. Then, is rank $(A + B) \leq$ rank (A) + rank (B)?

(e) For any $a, b, c, d, e, f, g, h, i \in R$, let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ and $B = \begin{bmatrix} e & d & f \\ b & a & c \\ h & g & i \end{bmatrix}$. Do A and B have the same value of determinant?

(f) If M and N are two subspaces of a vector space V and if every vector in V belongs either to M or to N (or both), then is $M = V$ or $N = V$ (or both)?

(g) Let A, B, C, D be square matrices of the same size, and A and D are of full rank. Then, is $\det.\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \det.\begin{bmatrix} D & C \\ B & A \end{bmatrix}$?

(h) Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}$.

(i) For a non-singular matrix A, if PAQ is an identity matrix, find A^{-1}.

(j) In a given system of linear equations $Ax = b$, if rank (A) = rank $(A : b)$ = number of unknown variables, what can you say about the system?

(k) When is a linear transformation $y = Ax$ said to be orthogonal? Indicate and usefulness of such a transformation.

(l) Is the matrix $A = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 2 & 3 & 1 \end{bmatrix}$ non-singular?

(m) Find the largest eigen value of the matrix $A = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$.

(n) When is a set of n $p \times 1$ vectors said to be linearly independent?

(o) When is a quadratic form $x'Ax$ said to be non-positive definite?

2. Answer ***any four*** questions : 5×4

(a) Find the condition to be satisfied by a matrix whose characteristic equation is $\lambda^2 - \lambda - 1 = 0$.

(b) If A, B and C are any three matrices such that the products considered below are defined, show that rank (AB) + rank $(BC) \leq$ rank (B) + rank (ABC).

(c) Let A and B be two symmetric matrices of the same order. Show that if $AB = BA$, then A and B can be simultaneously diagonalised.

(d) If U and V be subspaces of the n-dimensional Euclidean space, then show that rank $(U + V)$ = rank (U) + rank (V) − rank $(U \cap V)$.

(e) Let A and B be two symmetric matrices of which B is positive definite and A has eigen values either 0 or 1. Show that AB is an idempotent matrix.

(f) Let A be a positive definite matrix. Show that the quadratic form $x'Ax$ is strictly convex in x.

3. Answer ***any two*** questions : 10×2

(a) (i) Let S be a set of linearly independent vectors from a vector space V. Show that vectors in any subset of S are also linearly independent.

(ii) Show that every linearly independent set of vectors $B = \{v_1, v_2, \ldots\ldots, v_n\}$ from a finite dimensional vector space V can be extended to a basis of V.

(iii) Let $(v_1, v_2, \ldots\ldots, v_q)$ form a basis of a vector space V. Let $v \in V$. Show that the representation of v in terms of the basis vectors is unique.

(b) Consider a $n \times n$ matrix A.

(i) When is A said to be an orthogonal matrix?

(ii) Show that A is an orthogonal matrix if and only if the rows of A form an orthonormal basis of the n-dimensional Euclidean space.

(iii) Let $S = (v_1, v_2, ..., v_k)$ be a set of linearly independent vectors from a vector space V of dimension $n > k$. If a vector v is orthogonal to every vector in S, show that $(v_1, v_2, ..., v_k, v)$ forms a set of linearly independent vectors.

(c) (i) Let A be a $n \times n$ upper triangular matrix. Show that the determinant of the matrix is the product of the diagonal elements of A.

(ii) Suppose A and B are two square matrices such that A and $A + B$ are invertible. Show that $A^{-1} - (A + B)^{-1} = (I + A^{-1}B)^{-1}A^{-1}BA^{-1}$, where I denotes the identity matrix.